

RISK & RETURN IN INVESTING

Financial Mathematics • CA-III • Unit V and Unit VI

Name: _____

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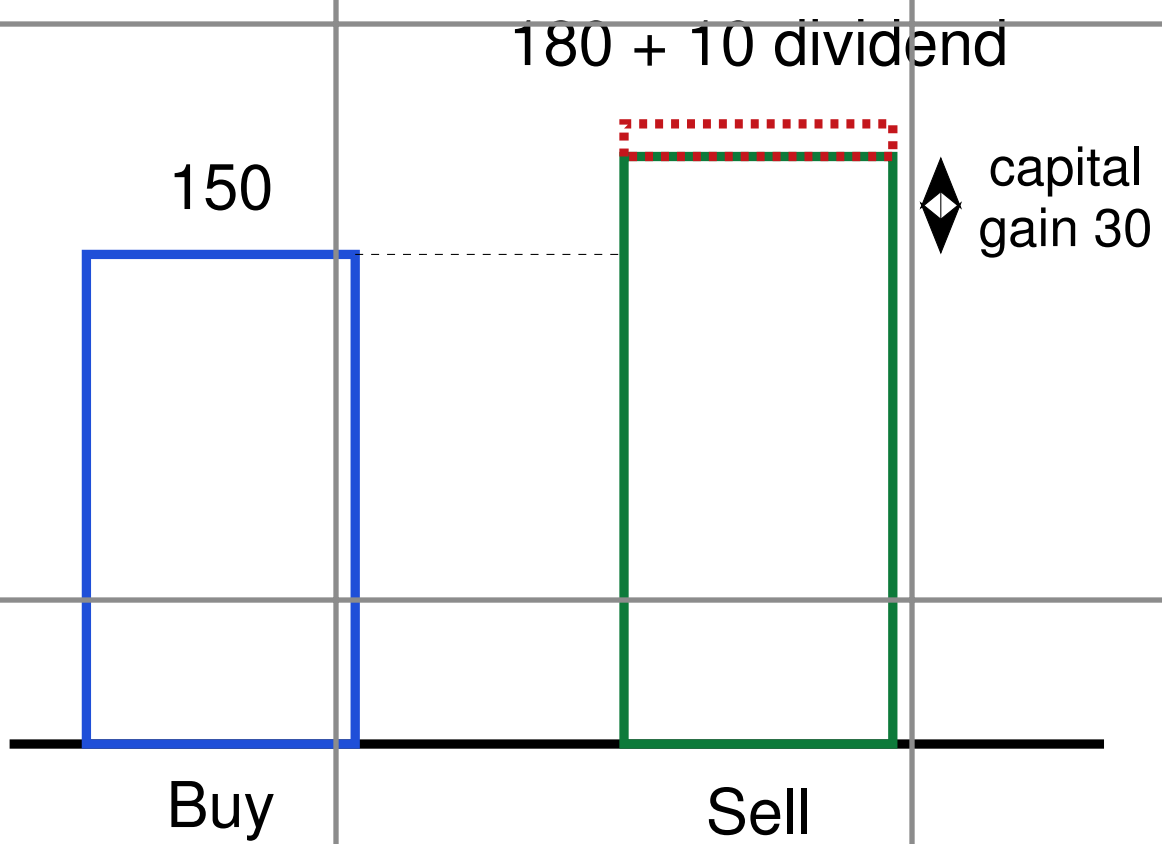
UNIT V: BASICS OF INVESTMENT

1. Investment Return

Gain or loss as a % of money invested.

$$\begin{aligned} \text{Absolute return} &= (V_f - V_i) / V_i \times 100 \\ \text{HPR} &= [\text{Income} + (V_f - V_i)] / V_i \times 100 \\ \text{CAGR} &= (V_f / V_i)^{1/n} - 1 \quad (\times 100 \text{ for } \%) \end{aligned}$$

Eg: Rs 150 → Rs 180, dividend Rs 10
HPR = (10 + 30) ÷ 150 = 26.67%



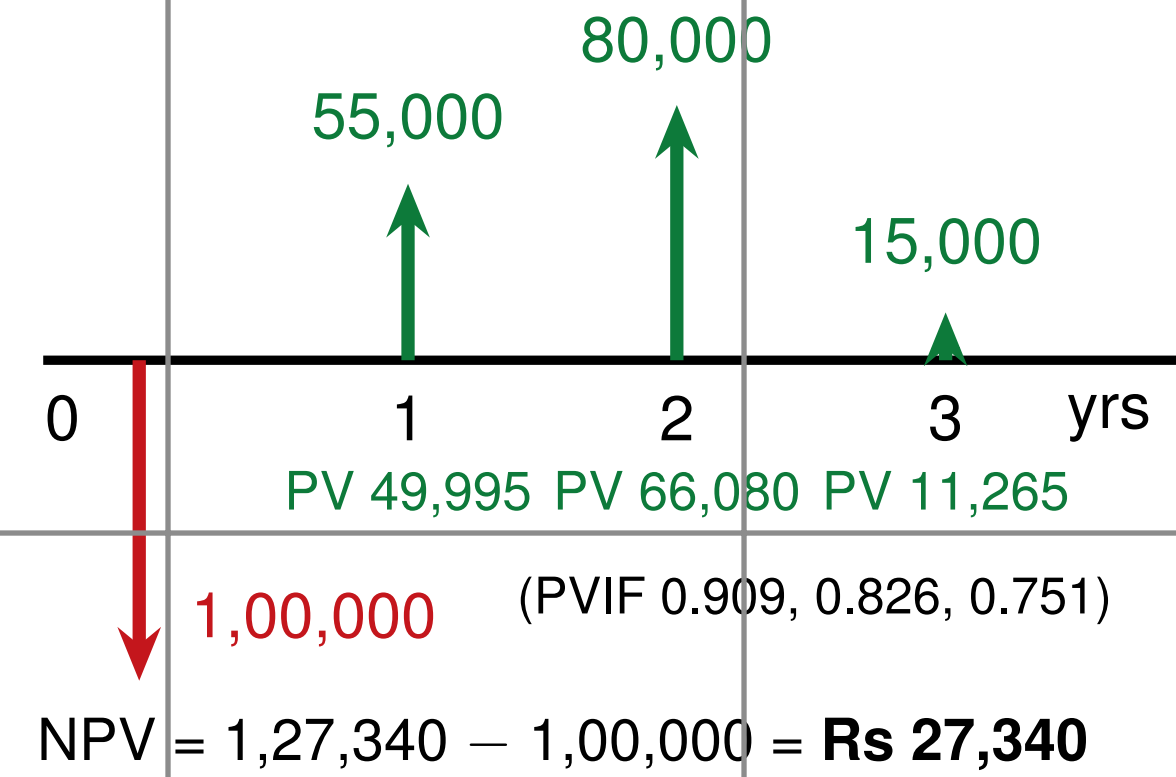
2. Uneven Cash Flows

Uneven flows: discount each to today.

$$NPV = \sum \frac{C_t}{(1+k)^t} - I$$

C_t = cash flow, k = rate, I = investment
 $NPV > 0$: accept $NPV < 0$: reject
 IRR = rate where $NPV = 0$

Eg: I = Rs 1,00,000, k = 10%

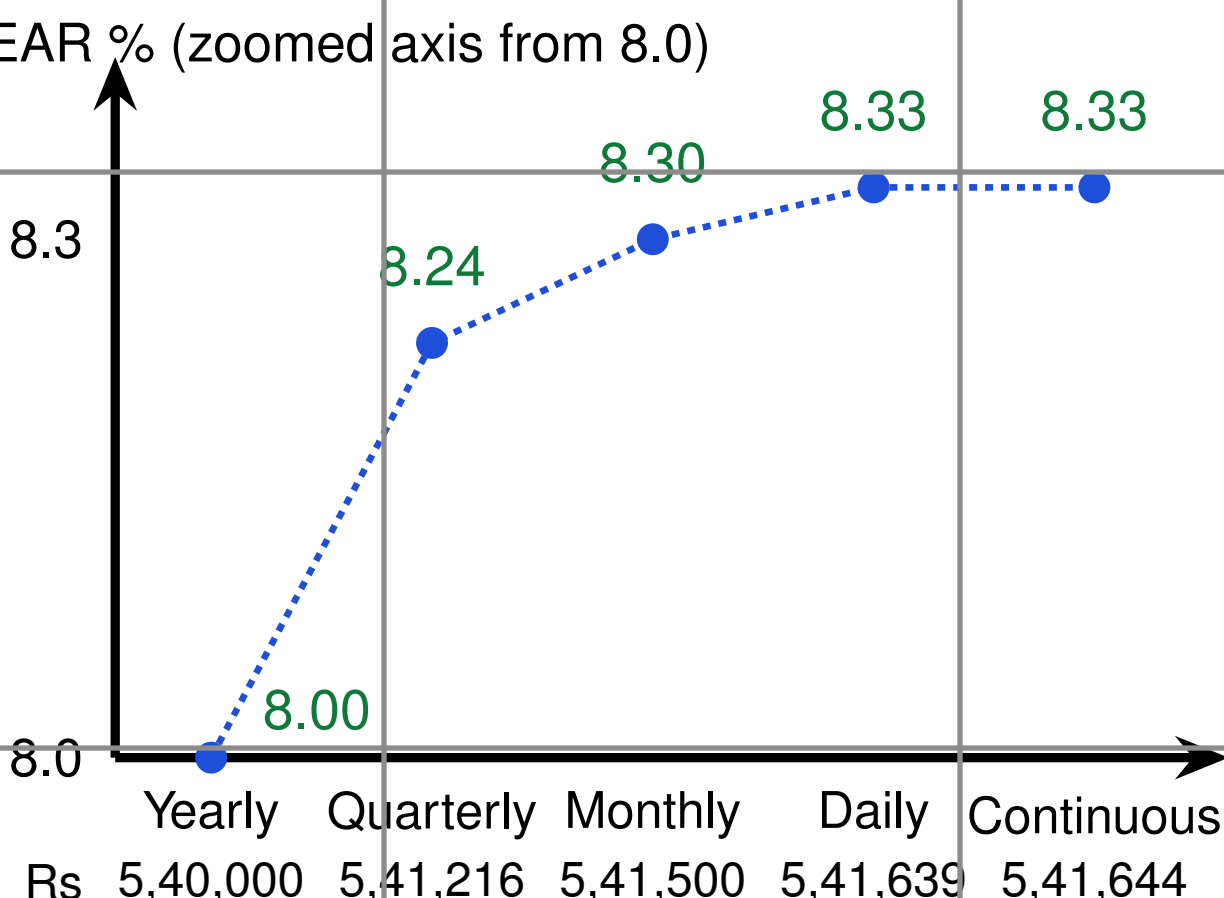


3. Compounding Frequency

More frequent = slightly more money.

$$\begin{aligned} A &= P \left(1 + \frac{r}{n}\right)^{nt} \quad \text{Continuous: } A = Pe^{rt} \\ \text{Effective annual rate} &= \left(1 + \frac{r}{n}\right)^n - 1 \end{aligned}$$

Eg: Rs 5,00,000 at 8%, 1 year

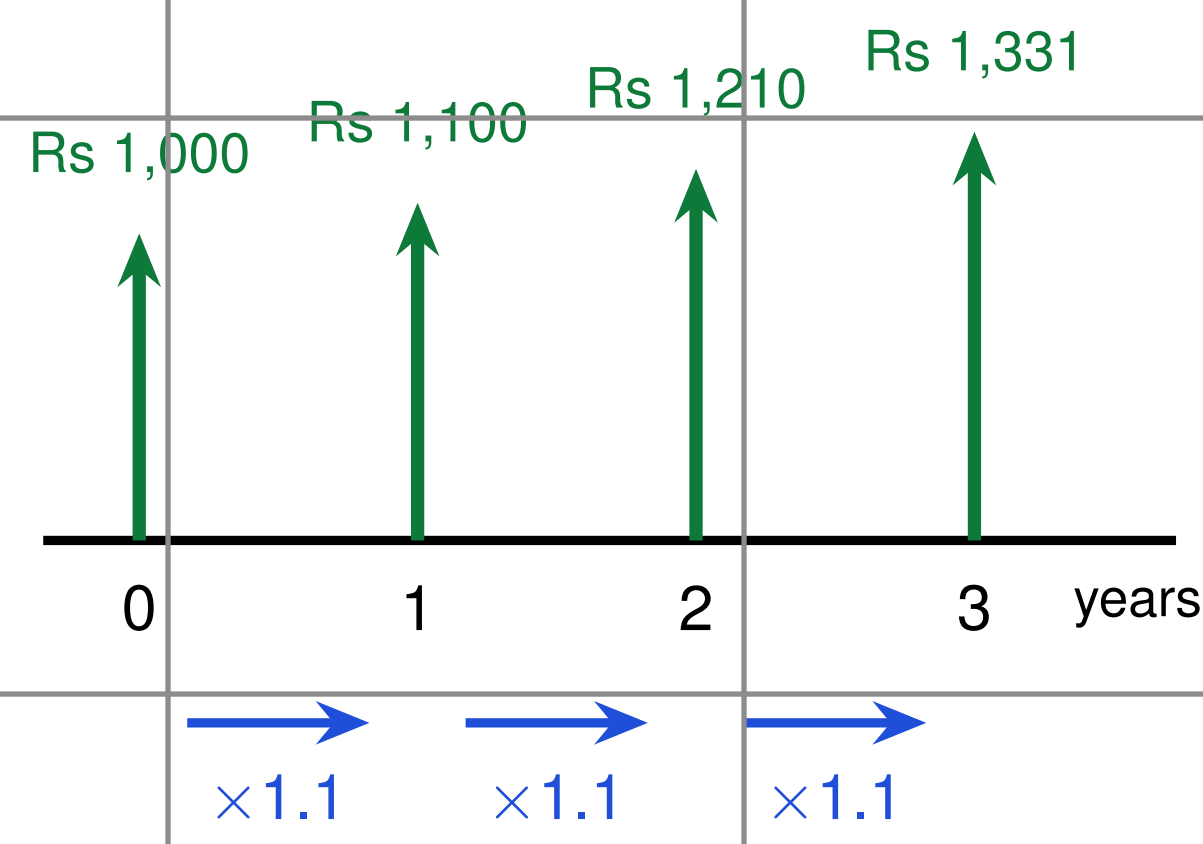


4. Economic Equivalence

Same value at a common point in time.

$$\begin{aligned} FV &= PV(1+i)^n \\ PV &= \frac{FV}{(1+i)^n} \end{aligned}$$

Eg: at 10%, Rs 1,000 today $\equiv$ Rs 1,331 after 3 years



UNIT VI: RISK AND UNCERTAINTY

1. Risk and Uncertainty

Risk = chance of an undesirable event that causes a financial loss.
Risk management = identify, assess, control.

Strategic

Business plan fails.
Kodak and digital photos

Compliance

Not following rules or laws

Operational

Mistakes by people: Rs 1,00,000 paid, not Rs 10,000

Financial

Prices, interest rates, exchange rates

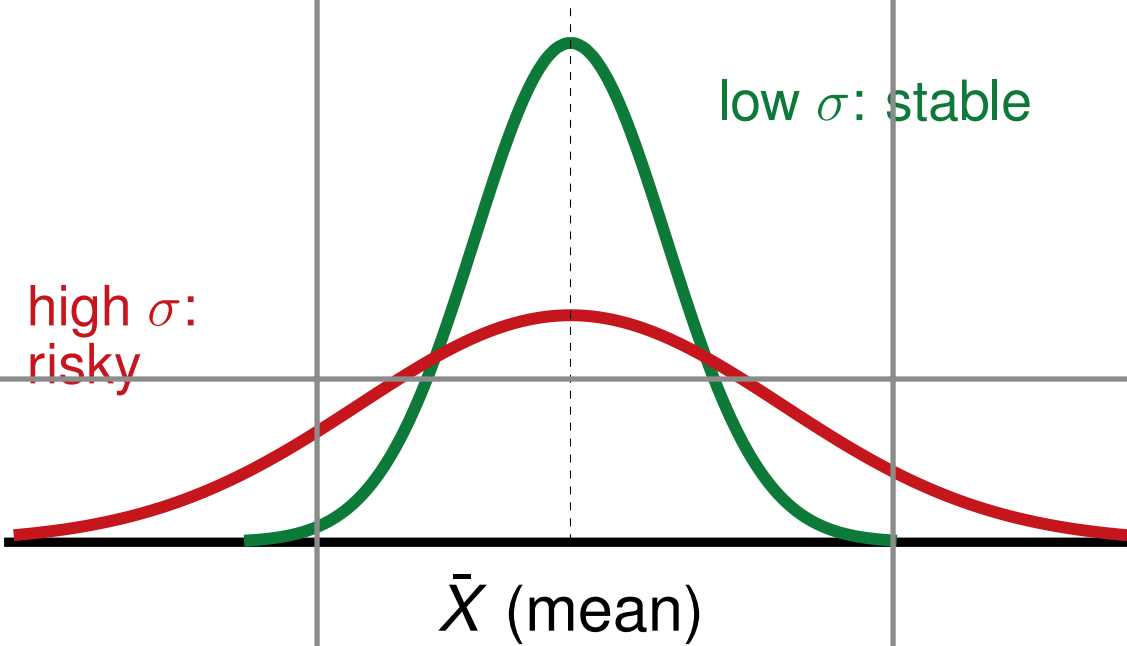
Risk

2. Measuring Risk

Risk is measured by SD and CV.

$$\begin{aligned} \text{Expected return } \bar{X} &= \sum X_i P(X_i) \\ \text{Variance } \sigma^2 &= \sum d^2 / n \quad (\text{or } \sum d^2 P) \\ \text{SD } \sigma &= \sqrt{\sigma^2}, \quad d = X_i - \bar{X} \\ \text{CV} &= \sigma / \bar{X} \times 100 \end{aligned}$$

Eg: 12, 18, 20, 15, 25 (%)
mean 18%, σ = 4.43% ($\div n$)
CV = 24.6%

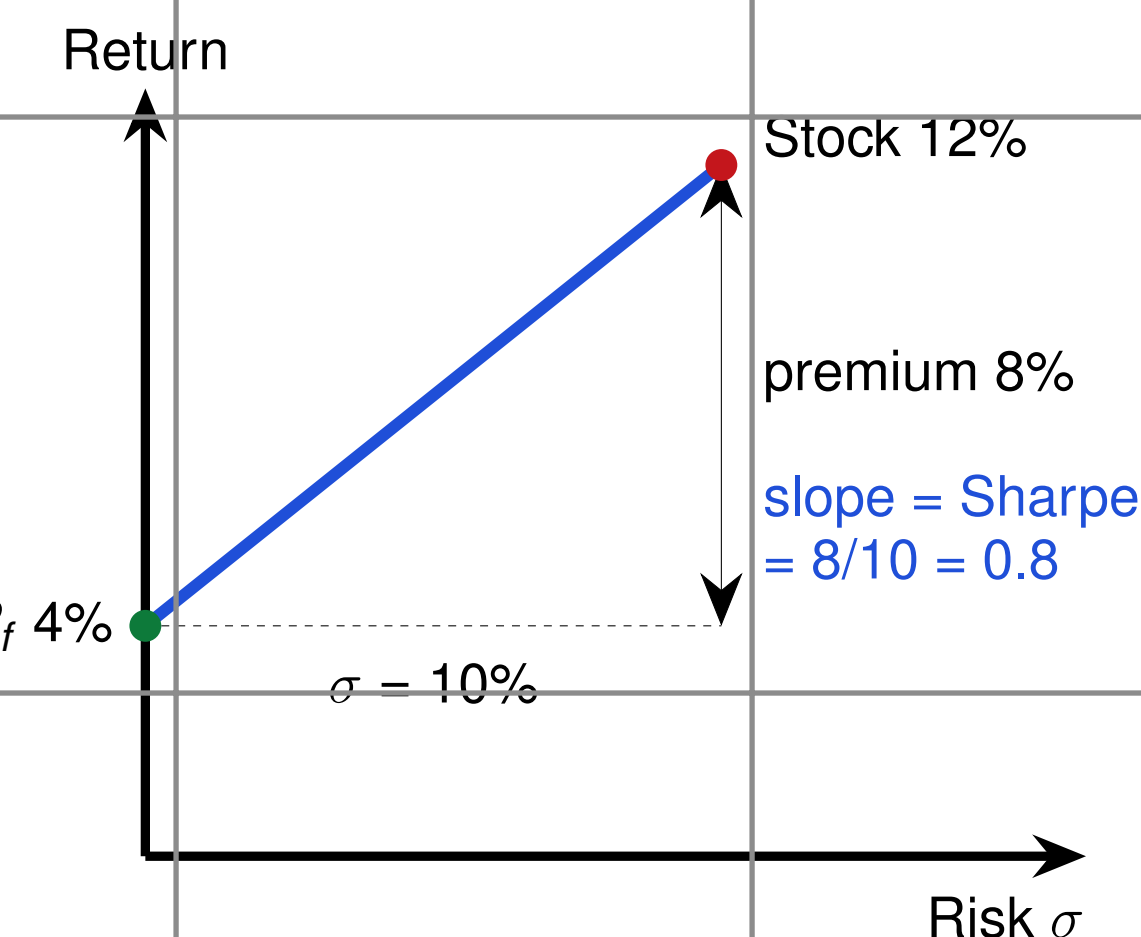


3. Risk Premium

Extra return for taking risk.

$$\begin{aligned} \text{Risk premium} &= E(R) - R_f \\ \text{Sharpe ratio} &= (R_p - R_f) / \sigma_p \end{aligned}$$

Eg: $E(R)$ = 12%, R_f = 4%, σ = 10%
Premium = 8%, Sharpe = 0.8

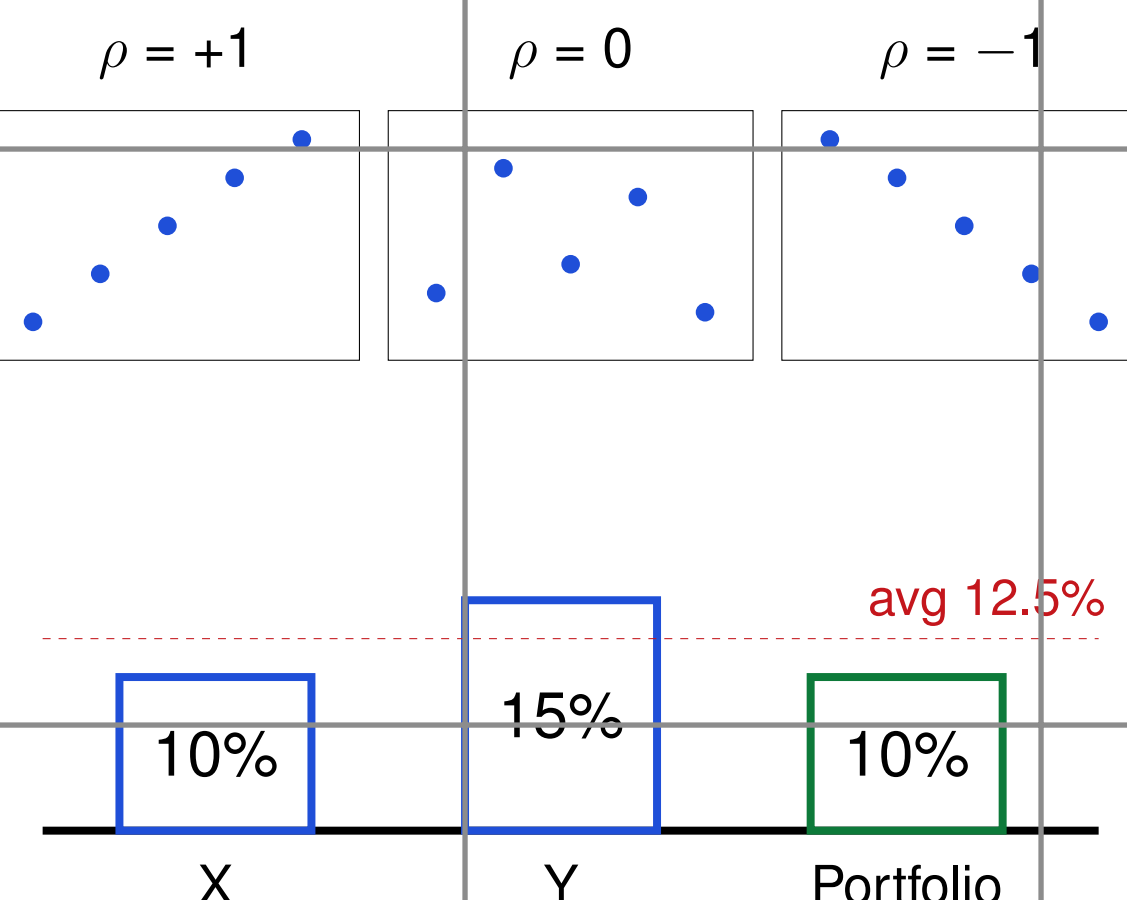


4. Portfolio Diversification

With $\rho < 1$: $\sigma_p < \text{weighted average}$.

$$\begin{aligned} \sigma_p^2 &= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 \\ &\quad + 2 w_1 w_2 \sigma_1 \sigma_2 \rho_{12} \end{aligned}$$

Eg: w = 0.5, 0.5; ρ = 0.25
 σ = 10%, 15%: σ_p = 10%

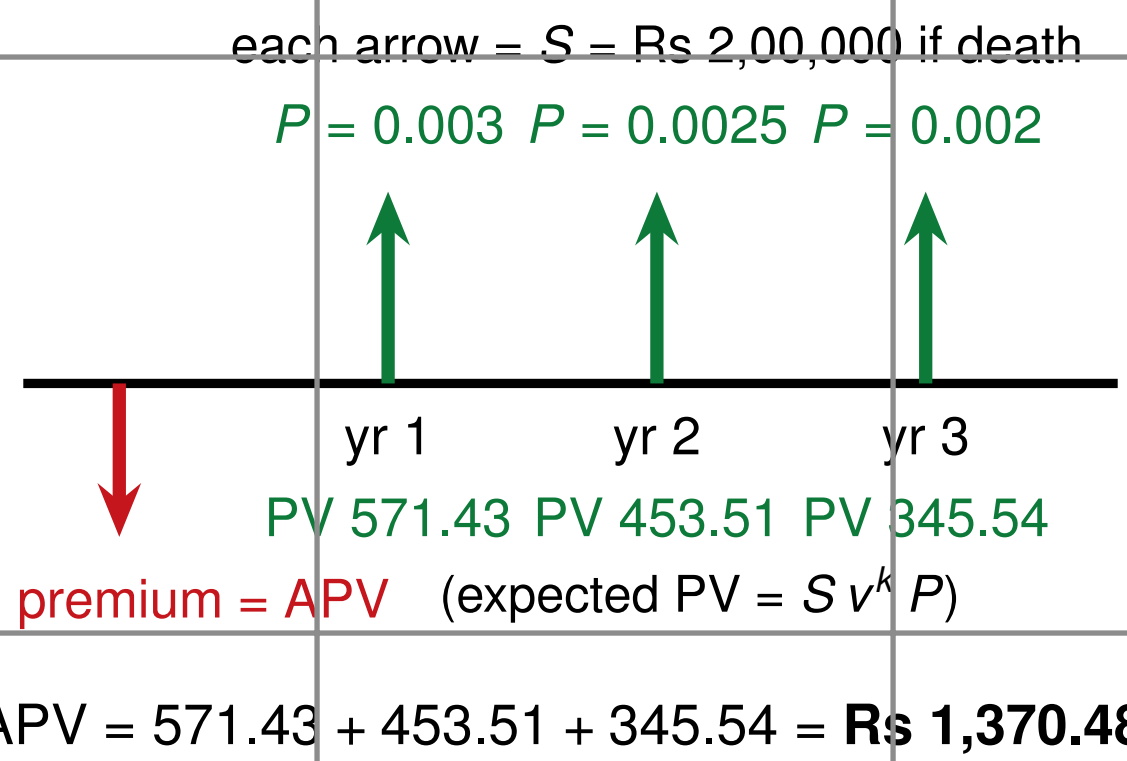


5. Life Insurance

Insurer pays sum assured S on death in the term.

$$\begin{aligned} \text{APV} &= S \sum_{k=1}^n v^k P(T = k) \\ v &= \frac{1}{1+i} \end{aligned}$$

Eg: 3-yr term, S = Rs 2,00,000, i = 5%
 $P(T = 1, 2, 3)$ = 0.003, 0.0025, 0.002



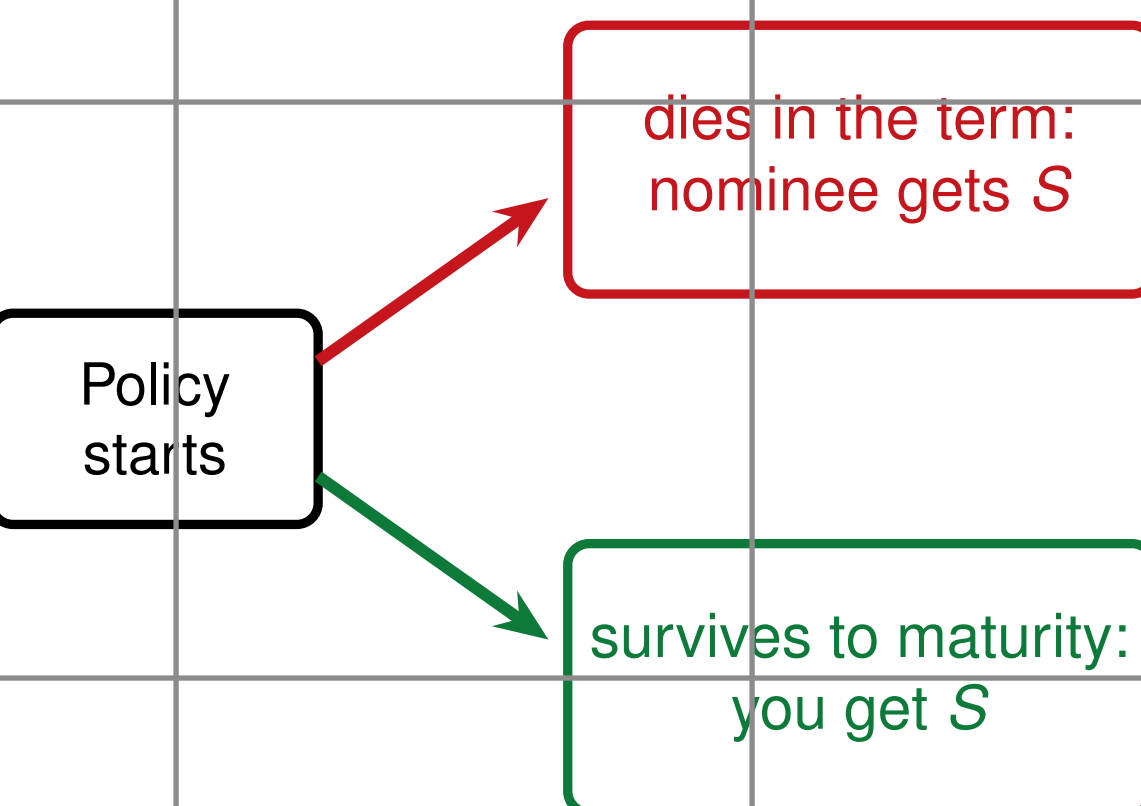
6. Endowment

Pays S if you die in the term OR survive to maturity.

$$\text{APV} = S \left[\sum_{k=1}^n v^k P(T = k) + v^n {}_n P_x \right]$$

${}_n P_x$ = chance of surviving n years

Eg: 20-yr plan, S = Rs 1,00,000



Investors demand a higher expected return for taking higher risk